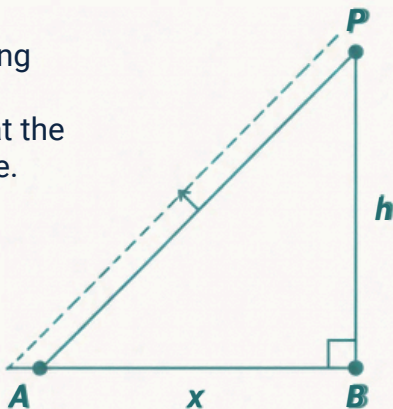

What geometry can tell us when measurements are uncertain

Research notes on height estimation, exact reformulation, and stronger benchmarks.

I've been building and testing these ideas privately. Here is one result, and what the next comparison taught me.

Ayokunle Olufosoye



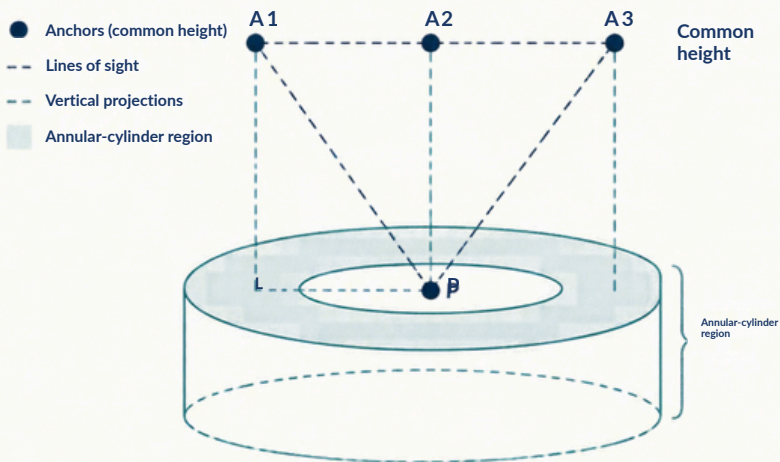
The question and the model

Question

How large can unavoidable height estimation error be when squared-distance readings have bounded errors?

Model

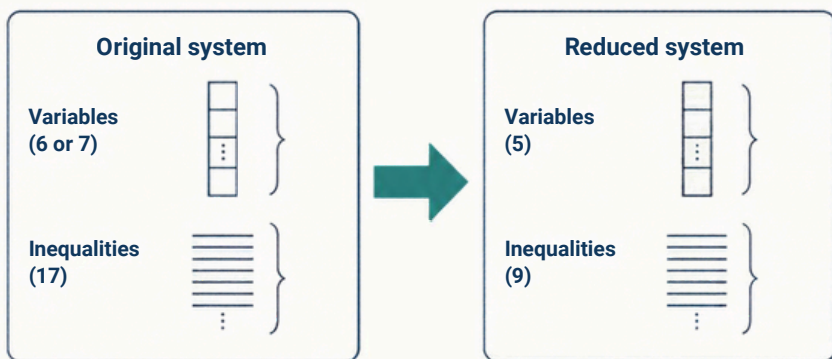
Known annular-cylinder region; three noncollinear anchors at a common height at or above the region; bounded shared bias in squared readings plus separately bounded residual errors.



Illustrative geometry only – not experimental data.

From ambiguity to a smaller optimization

Two physically admissible points can explain the same observations; the largest compatible height separation divided by two gives global minimax absolute height error.



Exact model-specific reformulation

Original: 6/7 variables · 17 inequalities

Reduced: 5 variables · 9 inequalities



Preserves the optimum under stated assumptions.



Priority/novelty not established.

Same solver, different formulation

52–239×
faster search

34–130×
fewer handled
search cells

Native IBEX

- original

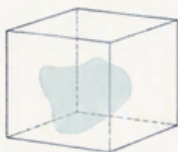
versus

reduced
formulation •

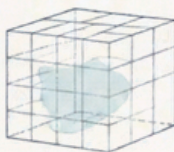
- SIX layouts •

matched settings

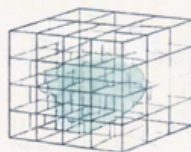
Search region



Cell subdivision



Finer cells near boundary



METHODS

**24 settings/layout
combinations**

**120 measured trials
plus 24 warmups**



SCOPE CAVEATS



Search time only; not end-to-end
application speed.



This comparison concerns
formulation within IBEX,
not heuristic versus IBEX.

A stronger baseline narrowed the claim

SEPARATE FOLLOW-UP COMPARISON

Compiled six-start SLSQP on original formulation versus reduced IBEX.



KEY RESULT (QUALIFIED)



26 UNIQUE layouts



Heuristic met the 1% accuracy target on all 26



At least 2x faster on 22 of 26

CANDIDATE AND BOUND



Feasible candidates give lower bounds.
Certification also bounds the optimum above.



SCOPE CAVEAT

Six starts selected using these layouts; NOT independent holdouts.



DIFFERENT GUARANTEES

Heuristic candidate versus interval-certified bound.



Separate follow-up — do not mix sample counts or speedup claims from the previous page.

What the evidence supports

1

The reformulation substantially accelerated certified search on this tested model.



2

The experiments did not establish a certification-cost advantage over a strong approximate method.



Open

broader models, application value, external review and novelty.

AI-assisted investigation and implementation.
Internal checks are not external peer review.

Where would a guaranteed bound change the decision you can make?

Where this could go next

1

Build a library of geometric reductions, each with explicit conditions for valid use.



2

Use constraint graphs and Bayesian selection to suggest reductions; verify each before use.



Potential uses

Localization, metrology and robot workspace verification.

Test on unseen geometries: total runtime, validity and preserved guarantees.

My next question: can this extend beyond one geometry?